

Supplementary Materials

1 Detailed Experiment Settings

Statistics	Agriculture	Art	Legal	Mix	News
Total Tokens	1,923,158	3,581,243	4,719,597	602,561	1,409,743
Total Chunks	1,838	3,414	5,424	596	1,614

Table 1: The detailed statistics of the five datasets.

Datasets Statistics In the QFS task, we choose a total of five datasets for evaluating the FG-RAG. The total number of tokens and the number of chunks partitioned into each dataset are summarized in Table 1.

In the multi-hop QA task, we choose the News dataset with multi-hop QA. Given the extensive collection of QA pairs in the multi-hop QA, we opted for a random selection of 125 pairs from each category—namely, inference, comparison, and temporal queries—for our evaluation process. The details of the three types of queries are described below:

- **Inference query** These queries are developed by integrating the various descriptions of the bridge-entity from multiple claims, ultimately leading to the identification of the entity itself.
- **Comparison query** These queries are crafted to highlight the similarities and differences concerning the bridge entity or topic, with the answer usually being a clear “yes” or “no” based on the comparative analysis.
- **Temporal query** These queries examine the sequence of events at different points in time, with answers typically being a “yes” or “no”, or a single temporal indicator such as “before” or “after”.

Implementation Details The chunk size for all datasets is standardized at 1200 tokens, with a 100-token overlap between consecutive chunks. In addition, the gleaning parameter is fixed to 1 for all GraphRAG-based methods that need to extract entities and relationships from the dataset.

Evaluation Details In the QFS task, we use the evaluation method presented in LightRAG [Guo *et al.*, 2024] to compare the answers given by different methods, and the LLM used in the evaluation process is gpt-4o-mini. We alternate the placement of the two answers in the prompt when comparing the answers given by two methods and calculate win rates accordingly. Specifically, we will treat the answer given

by a method as Answer 1 and Answer 2 respectively at one time, thus ensuring fair comparisons and minimizing possible bias due to the order in the prompt. In the multi-hop QA task, we judge an answer to be correct when and only when it is identical to the standard answer ignoring case.

2 Detailed Experiment Results

In this paper, when assessing the performance of the QFS task, we initially display the average performance of each model across all datasets. To gain a more detailed understanding of how FG-RAG’s performance compares with other baselines across various datasets, in Table 2 we present specific experimental results for each dataset. In addition to the baselines mentioned in the paper, we also made a comparison with FastGraphRAG. It is a graph-enhanced RAG system integrates graph structure with vector representation, utilizing the PageRank algorithm for graph retrieval to enhance the accuracy and reliability of information retrieval. However, due to the outputs of certain LLMs, like qwen2.5-7b-instruct, failing to meet FastGraphRAG’s strict formatting requirements, we are only able to get its response to the query when using gpt-4o-mini.

gpt-4o-mini	Agriculture		Art		Legal		Mix		News	
	NaiveRAG	FG-RAG	NaiveRAG	FG-RAG	NaiveRAG	FG-RAG	NaiveRAG	FG-RAG	NaiveRAG	FG-RAG
Comprehensiveness	37.2%	62.8%	36.0%	64.0%	33.2%	66.8%	33.2%	66.8%	38.0%	62.0%
Diversity	32.0%	68.0%	33.2%	66.8%	11.2%	88.8%	37.6%	62.4%	32.0%	68.0%
Empowerment	34.4%	65.6%	32.4%	67.6%	21.6%	78.4%	30.4%	69.6%	31.6%	68.4%
Overall	36.0%	64.0%	34.4%	65.6%	23.2%	76.8%	28.8%	71.2%	32.4%	67.6%
	LightRAG	FG-RAG	LightRAG	FG-RAG	LightRAG	FG-RAG	LightRAG	FG-RAG	LightRAG	FG-RAG
Comprehensiveness	46.0%	54.0%	40.4%	59.6%	47.6%	52.4%	39.2%	60.8%	45.2%	54.8%
Diversity	46.4%	53.6%	40.0%	60.0%	38.4%	61.6%	45.6%	54.4%	43.2%	56.8%
Empowerment	42.0%	58.0%	40.8%	59.2%	45.2%	54.8%	35.6%	64.4%	41.2%	58.8%
Overall	42.0%	58.0%	40.0%	60.0%	44.8%	55.2%	36.0%	64.0%	44.0%	56.0%
	GraphRAG	FG-RAG	GraphRAG	FG-RAG	GraphRAG	FG-RAG	GraphRAG	FG-RAG	GraphRAG	FG-RAG
Comprehensiveness	39.2%	60.8%	36.4%	63.6%	40.8%	59.2%	41.2%	58.8%	40.4%	59.6%
Diversity	30.8%	69.2%	30.8%	69.2%	32.4%	67.6%	42.4%	57.6%	36.0%	64.0%
Empowerment	34.0%	66.0%	30.4%	69.6%	30.0%	70.0%	33.6%	66.4%	40.8%	59.2%
Overall	34.8%	65.2%	32.8%	67.2%	34.4%	65.6%	35.2%	64.8%	39.6%	60.4%
	FastGraphRAG	FG-RAG	FastGraphRAG	FG-RAG	FastGraphRAG	FG-RAG	FastGraphRAG	FG-RAG	FastGraphRAG	FG-RAG
Comprehensiveness	36.4%	63.6%	31.2%	68.8%	26.4%	73.6%	32.0%	68.0%	35.2%	64.8%
Diversity	30.4%	69.6%	27.6%	72.4%	24.4%	75.6%	30.8%	69.2%	33.6%	66.4%
Empowerment	32.8%	67.2%	27.6%	72.4%	21.2%	78.8%	27.6%	72.4%	28.4%	71.6%
Overall	33.2%	66.8%	28.8%	71.2%	22.4%	77.6%	27.2%	72.8%	30.4%	69.6%
qwen2.5-7b-instruct	Agriculture		Art		Legal		Mix		News	
	NaiveRAG	FG-RAG	NaiveRAG	FG-RAG	NaiveRAG	FG-RAG	NaiveRAG	FG-RAG	NaiveRAG	FG-RAG
Comprehensiveness	44.8%	55.2%	37.6%	62.4%	18.8%	81.2%	32.0%	68.0%	28.4%	71.6%
Diversity	45.2%	54.8%	49.2%	50.8%	9.6%	90.4%	35.6%	64.4%	32.0%	68.0%
Empowerment	47.2%	52.8%	41.2%	58.8%	15.6%	84.4%	31.2%	68.8%	30.4%	69.6%
Overall	43.6%	56.4%	40.0%	60.0%	16.4%	83.6%	30.0%	70.0%	27.6%	72.4%
	LightRAG	FG-RAG	LightRAG	FG-RAG	LightRAG	FG-RAG	LightRAG	FG-RAG	LightRAG	FG-RAG
Comprehensiveness	46.4%	53.6%	31.2%	68.8%	40.4%	59.6%	41.2%	58.8%	29.6%	70.4%
Diversity	47.6%	52.4%	46.8%	53.2%	38.4%	61.6%	42.8%	57.2%	41.6%	58.4%
Empowerment	45.6%	54.4%	30.8%	69.2%	32.8%	67.2%	37.2%	62.8%	29.2%	70.8%
Overall	46.4%	53.6%	32.0%	68.0%	35.2%	64.8%	38.4%	61.6%	28.0%	72.0%
	GraphRAG	FG-RAG	GraphRAG	FG-RAG	GraphRAG	FG-RAG	GraphRAG	FG-RAG	GraphRAG	FG-RAG
Comprehensiveness	29.6%	70.4%	30.8%	69.2%	17.2%	82.8%	22.0%	78.0%	22.0%	78.0%
Diversity	40.8%	59.2%	41.6%	58.4%	25.6%	74.4%	29.2%	70.8%	27.6%	72.4%
Empowerment	18.8%	81.2%	25.6%	74.4%	13.2%	86.8%	16.8%	83.2%	18.4%	81.6%
Overall	24.8%	75.2%	29.6%	70.4%	15.2%	84.8%	19.6%	80.4%	20.8%	79.2%
qwen2.5-3b-instruct	Agriculture		Art		Legal		Mix		News	
	NaiveRAG	FG-RAG	NaiveRAG	FG-RAG	NaiveRAG	FG-RAG	NaiveRAG	FG-RAG	NaiveRAG	FG-RAG
Comprehensiveness	38.0%	62.0%	28.8%	71.2%	9.2%	90.8%	20.0%	80.0%	28.8%	71.2%
Diversity	34.4%	65.6%	31.6%	68.4%	3.2%	96.8%	18.8%	81.2%	19.2%	80.8%
Empowerment	34.0%	66.0%	26.8%	73.2%	6.4%	93.6%	16.8%	83.2%	24.4%	75.6%
Overall	34.8%	65.2%	25.6%	74.4%	6.4%	93.6%	16.4%	83.6%	25.6%	74.4%
	LightRAG	FG-RAG	LightRAG	FG-RAG	LightRAG	FG-RAG	LightRAG	FG-RAG	LightRAG	FG-RAG
Comprehensiveness	47.2%	52.8%	38.4%	61.6%	26.8%	73.2%	40.8%	59.2%	44.0%	56.0%
Diversity	42.4%	57.6%	38.0%	62.0%	21.6%	78.4%	40.0%	60.0%	50.4%	49.6%
Empowerment	44.4%	55.6%	33.6%	66.4%	21.2%	78.8%	32.4%	67.6%	40.8%	59.2%
Overall	46.0%	54.0%	36.4%	63.6%	23.6%	76.4%	35.6%	64.4%	43.2%	56.8%
	GraphRAG	FG-RAG	GraphRAG	FG-RAG	GraphRAG	FG-RAG	GraphRAG	FG-RAG	GraphRAG	FG-RAG
Comprehensiveness	28.0%	72.0%	28.2%	71.8%	23.6%	76.4%	14.4%	85.6%	22.0%	78.0%
Diversity	39.2%	60.8%	33.6%	66.4%	34.0%	66.0%	22.8%	77.2%	26.0%	74.0%
Empowerment	21.2%	78.8%	16.8%	83.2%	17.2%	82.8%	10.0%	90.0%	16.8%	83.2%
Overall	25.2%	74.8%	20.4%	79.6%	21.6%	78.4%	11.6%	88.4%	17.6%	82.4%
qwen2.5-1.5b-instruct	Agriculture		Art		Legal		Mix		News	
	NaiveRAG	FG-RAG	NaiveRAG	FG-RAG	NaiveRAG	FG-RAG	NaiveRAG	FG-RAG	NaiveRAG	FG-RAG
Comprehensiveness	44.0%	56.0%	36.0%	64.0%	35.2%	64.8%	44.4%	55.6%	31.6%	68.4%
Diversity	42.4%	57.6%	39.6%	60.4%	26.4%	73.6%	47.6%	52.4%	36.4%	63.6%
Empowerment	44.4%	55.6%	36.8%	63.2%	29.2%	70.8%	46.8%	53.2%	31.2%	68.8%
Overall	43.6%	56.4%	34.4%	65.6%	30.8%	69.2%	44.0%	56.0%	31.2%	68.8%
	LightRAG	FG-RAG	LightRAG	FG-RAG	LightRAG	FG-RAG	LightRAG	FG-RAG	LightRAG	FG-RAG
Comprehensiveness	42.4%	57.6%	31.6%	68.4%	31.6%	68.4%	38.4%	61.6%	32.4%	67.6%
Diversity	35.6%	64.4%	35.2%	64.8%	28.8%	71.2%	45.2%	54.8%	37.6%	62.4%
Empowerment	38.4%	61.6%	32.0%	68.0%	26.0%	74.0%	32.8%	67.2%	29.2%	70.8%
Overall	40.8%	59.2%	32.4%	67.6%	27.2%	72.8%	36.4%	63.6%	32.0%	68.0%
	GraphRAG	FG-RAG	GraphRAG	FG-RAG	GraphRAG	FG-RAG	GraphRAG	FG-RAG	GraphRAG	FG-RAG
Comprehensiveness	21.6%	78.4%	20.8%	79.2%	17.6%	82.4%	32.0%	68.0%	18.0%	82.0%
Diversity	36.0%	64.0%	30.8%	69.2%	35.2%	64.8%	47.6%	52.4%	26.0%	74.0%
Empowerment	19.2%	80.8%	16.8%	83.2%	16.4%	83.6%	28.4%	71.6%	16.0%	84.0%
Overall	21.6%	78.4%	21.2%	78.8%	17.6%	82.4%	33.6%	66.4%	16.0%	84.0%

Table 2: Comparison of FG-RAG's win rate with other baselines across LLMs, metrics, and datasets.

-Goal-
Given a passage of problem text, identify as many entities as possible. Output directly without explanation.

-Steps-
1. Identify as many entities as possible. For each identified entity, extract the following information:
- **entity_name**: Name of the entity.
2. Output all entities identified in step 1 separated by newlines. Output directly without explanation.

Example 1:
Text: What frameworks for character analysis can be employed to interpret the depth of character relationships presented?
Output:
frameworks
character analysis
depth
character relationships

Example 2:
Text: Who is the individual associated with the cryptocurrency industry facing a criminal trial on fraud and conspiracy charges?
Output:
individual
cryptocurrency industry
criminal trial
fraud charges
conspiracy charges

-Real Data-
Text: {input_query}
Output:

Query Entity Extraction

Figure 1: Prompts for query entities extraction.

-Goal-
Given a passage of problem text, identify as many entities as possible. Output directly without explanation.

-Steps-
1. Identify as many entities as possible. For each identified entity, extract the following information:
- **entity_name**: Name of the entity.
2. Output all entities identified in step 1 separated by newlines. Output directly without explanation.

.....

-Real Data-
Text: {input_query}
Output:

Query Entity Extraction

What do you need to know about the {entity_name} to fully answer this question? Just give what you want to know in the form of up to {max_question_number} questions. Ask directly without explanation.

Question Formulation

Figure 2: Prompts for relevant questions formulation.

-Goal-
Generate a response of the target format that around the entity {entity_name}, summarizing all given information for the response format, with the response being as relevant as possible to the given questions, and incorporating any relevant general knowledge. Do not include information where the supporting evidence for it is not provided.

-Target response format-
Multiple Paragraphs

-Given infomation-
{input_descriptions}

-Given questions-
{input_questions}

Add sections and commentary to the response as appropriate for the format. Style the response in markdown.

Summaries Generation

Figure 3: Prompts for summaries generation.

3 Overview of the Prompts Used in FG-RAG

3.1 Prompts for query entities extraction

The prompt in Figure 1 instructs the LLM to extract entities from the query. To prevent the LLM from overlooking key entities in queries during entity extraction, we apply few-shot prompting.

3.2 Prompts for relevant questions formulation

The prompt in Figure 2 instructs the LLM to ask relevant questions around a particular entity. Assuming that the query entity extracted in the entities extraction phase are $\{E_1, E_2, \dots, E_n\}$, we direct the LLM to ask a certain number of questions around the query entity E_i based on the context of the query entity extraction step.

3.3 Prompts for summaries generation

The prompt in Figure 3 instructs the LLM to generate a summary based on the retrieved content around a particular entity and some questions related to the entity. The generated summary consists of multiple paragraphs and is output as mark-down in sections.

3.4 Other prompts

Prompts used in other steps such as extracting entities and their relationships from text chunks, QFS task question generation, and evaluation of answer pairs are similar to those used in LightRAG [Guo *et al.*, 2024] and are not repeated here.

4 Case Studies Comparing FG-RAG with Other Baselines

In order to visualize the superiority of FG-RAG over other baselines in terms of comprehensiveness, diversity, and empowerment, we present some case studies. Table 3, Table 4, Table 5, and Table 6 show the answers given by FG-RAG compared to NaiveRAG [Gao *et al.*, 2023], LightRAG [Guo *et al.*, 2024], GraphRAG [Edge *et al.*, 2024], and Fast-GraphRAG in solving a complex query about how artistic crossovers affect political and social relations, respectively.

References

- [Edge *et al.*, 2024] Darren Edge, Ha Trinh, Newman Cheng, Joshua Bradley, Alex Chao, Apurva Mody, Steven Truitt, and Jonathan Larson. From local to global: A graph rag approach to query-focused summarization. *arXiv preprint arXiv:2404.16130*, 2024.
- [Gao *et al.*, 2023] Yunfan Gao, Yun Xiong, Xinyu Gao, Kangxiang Jia, Jinliu Pan, Yuxi Bi, Yi Dai, Jiawei Sun, and Haofen Wang. Retrieval-augmented generation for large language models: A survey. *arXiv preprint arXiv:2312.10997*, 2023.
- [Guo *et al.*, 2024] Zirui Guo, Lianghao Xia, Yanhua Yu, Tu Ao, and Chao Huang. Lightrag: Simple and fast retrieval-augmented generation. *arXiv preprint arXiv:2410.05779*, 2024.

Query	How have artistic crossovers influenced political and social relations historically?
FG-RAG	... The collaboration between diverse creative expressions has historically fostered dialogue around contemporary political climates and cultural dynamics. 1. Artistic Expression as a Catalyst for Change [...] 2. Public Art and Community Engagement [...] 3. The Role of Historical Figures [...] 4. Gender and Cultural Representation [...] Conclusion: In conclusion, the historical interplay between artistic crossovers and political/social relations ...
NaiveRAG	... It embodies cultural responses to political realities and social engagements. 1. The Role of Modernism [...] 2. The Shift to Postmodernism [...] 3. Cultural Politics and Resistance [...] Conclusion: Artistic crossovers have historically influenced political and social relations by enabling connections ...
LLM Decision	Comprehensiveness: “Winner”: “Answer 1 (FG-RAG)”, “Explanation”: “Answer 1 provides a detailed exploration of how artistic crossovers influence political and social relations, covering a variety of aspects ...” Diversity: “Winner”: “Answer 1 (FG-RAG)”, “Explanation”: “While both answers engage with multiple artistic movements, Answer 1 presents a wider range of perspectives by including diverse artistic forms, historical figures, and gender-based contributions. It highlights ...” Empowerment: “Winner”: “Answer 1 (FG-RAG)”, “Explanation”: “Answer 1 effectively empowers the reader by demonstrating how art can instigate discussions and change in society. It explains artistic crossovers and their implications for ...” Overall Winner: “Winner”: “Answer 1 (FG-RAG)”, “Explanation”: “Answer 1 is the overall winner due to its depth of detail, breadth of perspectives, and strong emphasis on empowerment. It effectively articulates the historical significance ...”

Table 3: A representative case comparing NaiveRAG to our FG-RAG method.

Query	How have artistic crossovers influenced political and social relations historically?
FG-RAG	... The collaboration between diverse creative expressions has historically fostered dialogue around contemporary political climates and cultural dynamics. 1. Artistic Expression as a Catalyst for Change [...] 2. Public Art and Community Engagement [...] 3. The Role of Historical Figures [...] 4. Gender and Cultural Representation [...] Conclusion: In conclusion, the historical interplay between artistic crossovers and political/social relations ...
LightRAG	... Below are some of the most informative metrics commonly used in this context, derived from the data and concepts surrounding recommendation systems. 1. Cultural Exchange and Its Political Implications [...] 2. Art as a Tool for Social Movements [...] 3. Reflecting and Shaping Societal Values [...] Conclusion: Artistic crossovers have historically played a significant role in ...
LLM Decision	Comprehensiveness: “Winner”: “Answer 1 (FG-RAG)”, “Explanation”: “Answer 1 provides a thorough examination of the role of artistic crossovers in political and social relations, encompassing various aspects such as historical examples, the ...” Diversity: “Winner”: “Answer 1 (FG-RAG)”, “Explanation”: “Answer 1 offers a greater variety of perspectives on the topic. It discusses not only the impact of individual artists but also artistic collaborations, community engagement ...” Empowerment: “Winner”: “Answer 1 (FG-RAG)”, “Explanation”: “Answer 1 empowers readers by providing them with a nuanced understanding of the subject, illustrating how art serves both as a mirror and a catalyst for societal change ...” Overall Winner: “Winner”: “Answer 1 (FG-RAG)”, “Explanation”: “Answer 1 is the overall winner as it excels in all three criteria: it is comprehensive in addressing various dimensions of the question, diverse in the perspectives and ...”

Table 4: A representative case comparing LightRAG to our FG-RAG method.

Query	How have artistic crossovers influenced political and social relations historically?
FG-RAG	... The collaboration between diverse creative expressions has historically fostered dialogue around contemporary political climates and cultural dynamics. 1. Artistic Expression as a Catalyst for Change [...] 2. Public Art and Community Engagement [...] 3. The Role of Historical Figures [...] 4. Gender and Cultural Representation [...] Conclusion: In conclusion, the historical interplay between artistic crossovers and political/social relations ...
GraphRAG	... These intersections can be observed across various historical contexts, illustrating how art has been utilized to advocate for social change, influence public perception, and foster diplomatic relations. 1. Art as a Catalyst for Social Change [...] 2. Political Commentary Through Artistic Expression [...] 3. Diplomatic Relations and Cultural Exchange [...] Conclusion: In summary, artistic crossovers have historically influenced political and social relations by serving as vehicles for social change, platforms for ...
LLM Decision	Comprehensiveness: “Winner”: “Answer 1 (FG-RAG)”, “Explanation”: “Answer 1 provides a detailed exploration of various aspects of how artistic crossovers influence political and social relations. It discusses specific artists and movements, such as Ibsen and Rubens, while also addressing ...” Diversity: “Winner”: “Answer 1 (FG-RAG)”, “Explanation”: “Answer 1 showcases a wider range of artistic forms and historical examples, including theatre, public art, and unique contributions from women artists. In contrast, Answer 2 focuses more heavily on ...” Empowerment: “Winner”: “Answer 1 (FG-RAG)”, “Explanation”: “Answer 1 effectively helps the reader understand the significant role of artistic expressions in societal and political contexts, encouraging them to reflect on the lasting impact of these artworks. It emphasizes ...” Overall Winner: “Winner”: “Answer 1 (FG-RAG)”, “Explanation”: “Overall, Answer 1 emerges as the stronger response due to its comprehensive detail, diverse range of examples and perspectives, and its ability to empower ...”

Table 5: A representative case comparing GraphRAG to our FG-RAG method.

Query	How have artistic crossovers influenced political and social relations historically?
FG-RAG	... The collaboration between diverse creative expressions has historically fostered dialogue around contemporary political climates and cultural dynamics. 1. Artistic Expression as a Catalyst for Change [...] 2. Public Art and Community Engagement [...] 3. The Role of Historical Figures [...] 4. Gender and Cultural Representation [...] Conclusion: In conclusion, the historical interplay between artistic crossovers and political/social relations ...
FastGraphRAG	... This phenomenon can be traced across different eras and movements, where art not only served as a medium of expression but also as a catalyst for change and reflection of societal dynamics. 1. Art as a Reflection of Political Change [...] 2. Influence of the Harlem Renaissance [...] 3. The Role of Baroque and Renaissance Art [...] 4. Postmodernism and Cultural Critique [...] Conclusion: In conclusion, the interplay between artistic crossover and political as well as social relations throughout history demonstrates the profound ...
LLM Decision	Comprehensiveness: “Winner”: “Answer 1 (FG-RAG)”, “Explanation”: “Answer 1 provides a detailed exploration of various aspects of how artistic crossovers influence political and social relations. It discusses specific artists and movements, such as Ibsen and Rubens, while also addressing ...” Diversity: “Winner”: “Answer 1 (FG-RAG)”, “Explanation”: “Answer 1 showcases a richer diversity of perspectives by incorporating a wider range of artistic examples, historical contexts, and gender considerations. It discusses the role ...” Empowerment: “Winner”: “Answer 1 (FG-RAG)”, “Explanation”: “Answer 1 empowers readers by offering them a deeper understanding of how art can reflect societal issues and incite change. It encourages critical thinking about the role ...” Overall Winner: “Winner”: “Answer 1 (FG-RAG)”, “Explanation”: “Answer 1 is the overall winner as it excels in all three criteria—comprehensiveness, diversity, and empowerment. It provides a thorough and nuanced exploration of the intersections between ...”

Table 6: A representative case comparing FastGraphRAG to our FG-RAG method.